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Palaeomagnetic data from a Mesozoic Philippine Sea Plate ophiolite on Obi Island, Eastern Indonesia

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Abstract

Palaeomagnetic data are presented from part of the Halmahera ophiolite exposed on Obi Island, eastern Indonesia. Until the late Neogene, Obi formed part of the southern Philippine Sea Plate; it is now isolated from that plate and is located between fault strands in the left-lateral Sorong Fault Zone. Two areas were sampled: the first area comprised two sites from a microgabbro and a third site in a thin intruding dyke, and the second area yielded one site from a sheeted dyke suite. The mean in situ direction for the two areas is $D = 216.1°$, $I = 23.3°$, where the angular separation is 34.7°. Rotating the mean directions back to the palaeo-vertical clusters the vectors, so that $D = 219.4°$, $I = 12.1°$, where the angular separation is 20.1°. This clustering, together with other lines of palaeomagnetic evidence, suggests that the magnetisation is primary. The ophiolite is Mesozoic, and most likely formed in the Jurassic. This information, together with recently published palaeomagnetic data from nearby Upper Cretaceous Philippine Sea Plate formations, suggest that the oldest parts of the Philippine Sea Plate were situated close to the equator in the western Pacific in the middle Mesozoic. © 2001 Elsevier Science Ltd. All rights reserved.

Keywords: Palaeomagnetic data; Mesozoic ophiolite; Philippine Sea Plate

1. Introduction

Until recently, palaeomagnetic data from eastern Indonesia were particularly scarce. This situation has been partially redressed following a concerted effort (254 sampled sites) in the major left-lateral Sorong Fault Zone system between New Guinea and Sulawesi (Fig. 1). Palaeomagnetically-based tectonic models for the region (Ali and Hall, 1995; Hall et al., 1995a–c; Hall, 1996) indicate that the Cenozoic tectonic history of eastern Indonesia and northern New Guinea has been dominated by the punctuated clockwise rotation of the Philippine Sea Plate and its interaction with the northward drifting Australia continental plate. Since the start of the Neogene, the Sorong strike-slip fault system has formed the boundary between the two plates. The relative motion of the two plates had led to the transfer of fragments, mainly from the Australian Plate to the Philippine Sea Plate, and the development of a broad fault zone in which fragments are partly coupled to the main plates.

Most of the published palaeomagnetic sites (Ali and Hall, 1995; Hall et al., 1995a–c) are principally from middle Paleogene and younger arc volcanic rocks and associated sediments which formed within the Philippine Sea Plate. Three Mesozoic formations, which are the sedimentary cover of ophiolitic basement and represent the oldest rocks forming part of the Philippine Sea Plate in the North Moluccas, also yielded directional information. Data from the Upper Cretaceous Gowonli and Gau Limestone Formations on eastern Halmahera (Hall et al., 1995a) and the Leleobasso Formation on NE Obi (Ali and Hall, 1995) indicate that the present-day southern Philippine Sea Plate was at sub-equatorial latitudes in the late Mesozoic. Early suggestions of large northward motions and rotation of the Philippine Sea Plate were based on magnetic anomaly studies and inclination data from ocean drilling (Louden, 1976, 1977; Keating, 1980; Keating and Herrero, 1980; Kinoshita, 1980; Bleil, 1981). These indicated a long-term northward translation of the plate, and this aspect of its motion history is now generally accepted.

However, if such unrotated northward motion of the Philippine Sea Plate is extended back to the late Cretaceous then Upper Cretaceous rocks of the Halmahera region should yield southern hemisphere palaeolatitudes of about

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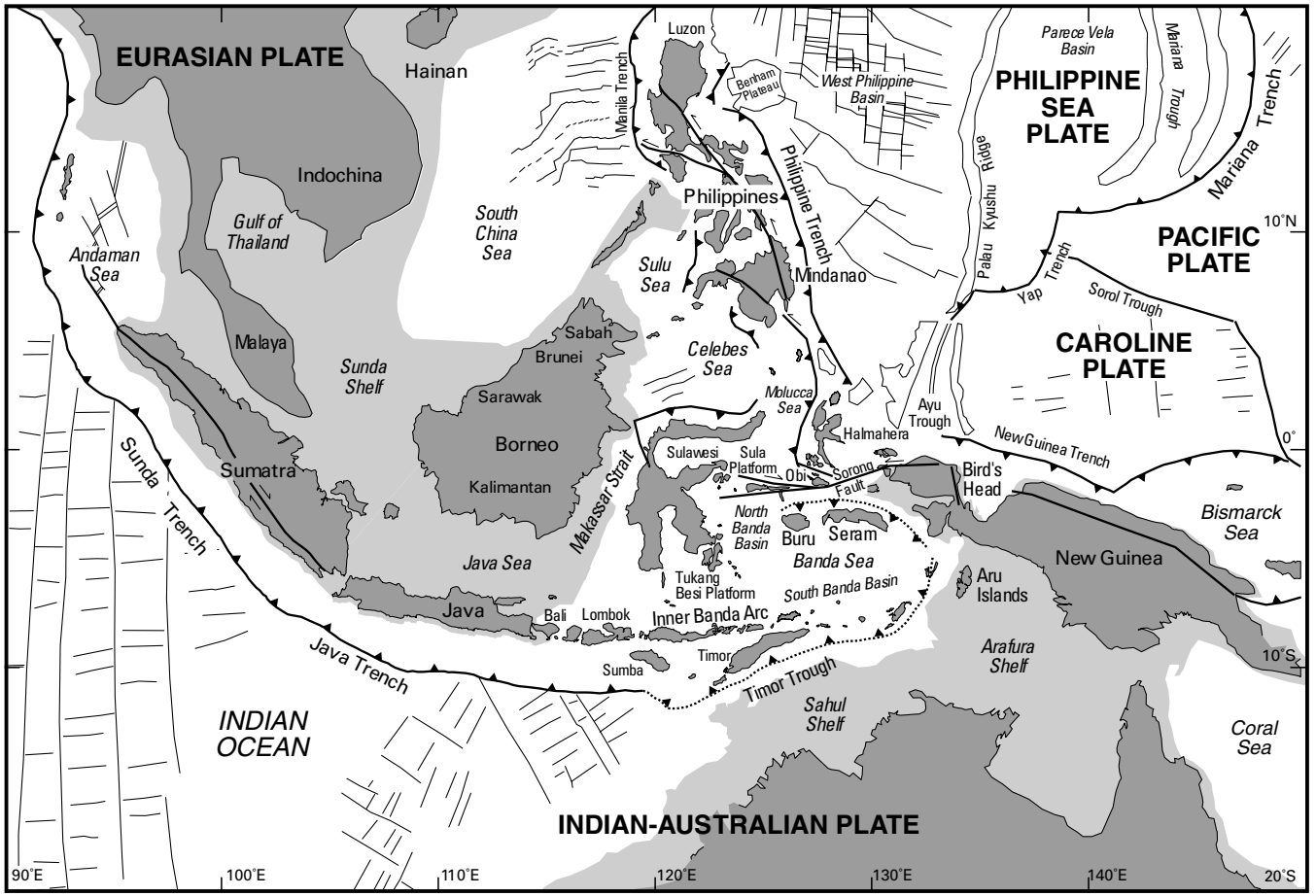


Fig. 1. Principal geographical features with major tectonic elements of SE Asia. The light shaded areas are the continental shelves of Eurasia and Australia drawn at the 200 m isobath.

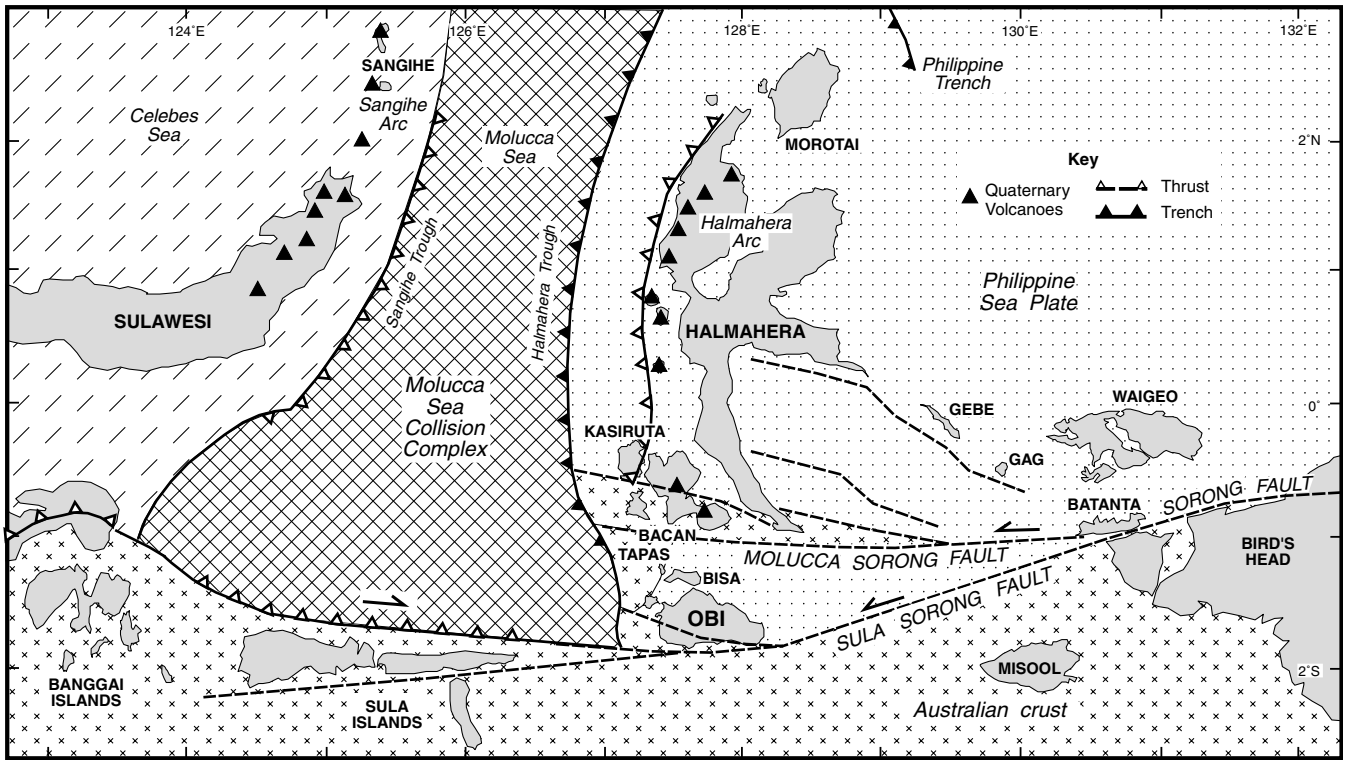


Fig. 2. The principal tectonic elements of the Sorong Fault Zone, east Indonesia and the location of islands of the Obi group.

30°S. They do not. One reason for this is that northward motion of the Philippine Sea Plate was accompanied by rotation of the plate (see Haston and Fuller, 1991; Koyama et al., 1992; Hall et al., 1995c).

Hall et al. (1995a–c) proposed that the Philippine Sea Plate had, since the middle Eocene, undergone three phases of clockwise rotation (50° between 50 and 40 Ma about an Euler pole at approximately 15°N, 160°E; 35° between 25 and 5 Ma about an Euler pole at approximately 10°N, 150°E; and 5° between 5 and 0 Ma about an Euler pole at 48.2°N, 157.0°E), with no rotation between 40 and 25 Ma. This model can account for almost all of the large latitudinal northward shifts and declination offsets reported in studies of the Philippine Sea Plate. The Upper Cretaceous palaeolatitudes are consistent with this rotation history since it predicts that the Halmahera region would have been at low latitudes in the late Cretaceous. However, the basement rocks of the North Moluccas in the Philippine Sea Plate are mainly ophiolites older than late Cretaceous. No palaeomagnetic information has previously been reported from these rocks which could be used to determine the Mesozoic position of the plate at times close to the time of origin of the basement ophiolites. We have since analysed data collected during field expeditions to the island of Obi (Fig. 2), within the Sorong Fault Zone, between 1990 and 1992, where these ophiolites were sampled. These data are discussed here.

2. Tectonic setting of eastern Indonesia

At the present-day, eastern Indonesia includes the junction between the Eurasian, Australian and Philippine Sea Plates (Hamilton, 1979) but in a very complex configuration. Most of the islands of the North Moluccas are within the extreme southern part of the Philippine Sea Plate (Figs. 1 and 2) which converges with Eurasia in the Philippines. North of Halmahera the Philippine Sea Plate is being subducted beneath the Philippines at the Philippine Trench but the trench terminates at about the latitude of Morotai. Eurasia–Philippine Sea Plates convergence is then distributed in a complex way in the Molucca Sea Collision Zone where the opposed Halmahera and Sangihe arcs are actively converging. The southern boundary of the Molucca Sea and the Philippine Sea Plate is the Sorong Fault system. The Sorong Fault extends through the northern Bird's Head region of New Guinea into several Pliocene-Recent left-lateral splays of the Sorong Fault. The island of Obi lies within this region of splays, south of Halmahera and west of the Bird's Head.

The Halmahera–Waigeo islands north of the Sorong Fault today form part of the Philippine Sea Plate and have a basement of ophiolitic and arc rocks (Hall et al., 1991). The ophiolites are remnants of an early Mesozoic intra-oceanic arc (Hall et al., 1988) and are overlain by Upper Cretaceous–Eocene arc volcanic and sedimentary rocks, and arc plutonic rocks intrude the ophiolites. All these

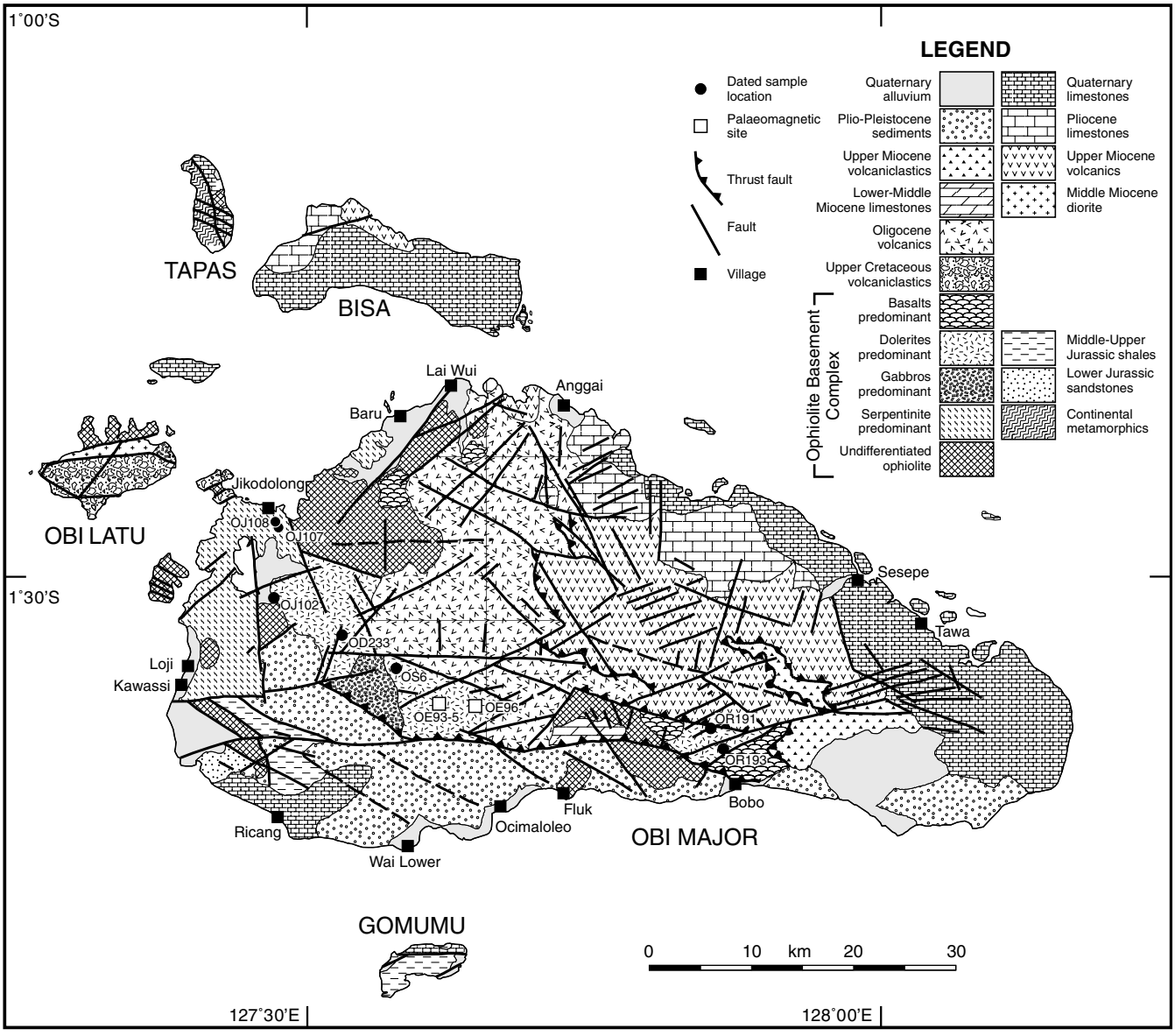


Fig. 3. Geological map of Obi based on SE Asia Research Group studies of the island and modified from Agustiyanto (1996).

older rocks are overlain by volcanic and sedimentary rocks formed principally during several later episodes of subduction-related volcanic activity. Volcanic activity related to Molucca Sea subduction continues at present in the northern part of the Halmahera arc.

South of the Sorong Fault there is crust of Australian origin (Visser and Hermes, 1962; Hamilton, 1979; Dow and Sukamto, 1984). The Bird's Head and Misool include passive continental margin sequences of Mesozoic and Cenozoic age and the oldest rocks known are lower Palaeozoic greywackes, which presumably overlie still older metamorphic basement rocks. The Sorong Fault cuts the Bird's Head and broadly separates these rocks from Eocene to Miocene island-arc sequences which resemble the Eocene–Oligocene arc rocks of the Halmahera region.

Within the strands of the Sorong Fault are areas of both ophiolitic/arc origin and continental crust and these are exposed on the islands between Bacan and Obi. They have been juxtaposed by left-lateral motion between splays of the Sorong Fault Zone since its inception in the early Miocene and different blocks have suffered variable local rotations within the fault zone (Ali and Hall, 1995).

3. Geology of Obi

Obi (Fig. 3) can be divided into two parts with different pre-Miocene geological histories based upon Dutch reconnaissance work (Wanner, 1913; Brouwer, 1924), mapping by GRDC (Sudana and Yasin, 1983) and our own studies (Hall et al., 1991; Ali and Hall, 1995; Agustiyanto, 1996). The major part of the island in the

north has a basement of Mesozoic ophiolitic rocks, Upper Cretaceous arc volcaniclastic sedimentary rocks of the Leleobasso Formation and the Oligocene Anggai River Formation arc volcanic and volcaniclastic rocks. Diorite plutons intrude the ophiolitic and Cretaceous rocks in west Obi. These rocks are equivalent to similar but more complete sequences of Halmahera and Waigeo and have a Philippine Sea Plate origin (Hall et al., 1995a).

The south-western part of the island is underlain by Australian-origin continental rocks. Continental metamorphic rocks probably form the basement in SW Obi where they are found as float samples in rivers and there is a sequence of Mesozoic sedimentary rocks, the Soligi and Gomumu Formations, unlike any of the Philippine Sea Plate rocks. The Soligi Formation comprises Lower Jurassic sandstones and siltstones with fragments of *Pentacrinus*. In SW Obi and on the small island of Gomumu to the south there are siltstones and shales of the Gomumu Formation. This formation locally contains a rich fauna including ammonite fragments, aptychi, belemnites and bivalves. Palynomorphs and belemnites indicate middle–Upper Jurassic ages. Wanner (1913) reported Jurassic ammonites as float in SW Obi. The Soligi and Gomumu Formations are closely similar to Jurassic rocks of the Australian margin known throughout eastern Indonesia.

On the islands of Bisa and Tapas, immediately NW of Obi, are high grade metamorphic rocks similar to those exposed on Bacan 50 km to the north. Metabasic rocks in this complex yield radiometric ages >100 Ma (Baker, 1997; Malaihollo, 1993; Malaihollo and Hall, 1996) and are probably deep arc crust from the Philippine Sea Plate. Continental metamorphic rocks on Bisa, Tapas and Bacan, including garnet-kyanite schists and gneisses, are presumed to be Palaeozoic or older and to be derived from the Australian continental margin. Isotopic dating of these rocks from Tapas and Bacan had yielded very young ages which are reset by Neogene volcanic and hydrothermal activity (Baker and Malaihollo, 1996).

The major part of Obi is overlain locally by Miocene shallow water limestones and then by a thick sequence of middle–Upper Miocene arc volcanic and volcaniclastic rocks of the Woi formation and their equivalent marine forearc deposits of the Guyuti Formation. These are the oldest products of the Halmahera volcanic arc and are overlain in north Obi by Pliocene limestones and in south Obi by Plio-Pleistocene conglomerates and sandstones. There is a Neogene diorite body on Obi Latu of similar age to the volcanic rocks on Obi.

4. Philippine Sea Plate ophiolite basement

Much of our knowledge of the Philippine Sea Plate basement rocks within the Sorong Fault Zone has resulted from the work of the SE Asia Research Group (Hall et al., 1988, 1991; Ballantyne, 1990, 1991, 1992; Agustiyanto, 1996; Baker, 1997). Ballantyne established that the ophiolite basement formed within a supra-subduction zone setting. The main body of this ophiolite is exposed in east Halmahera, but in Gag, Gebe and Waigeo there are parts of the same ophiolitic terrain, and on islands along the Sorong Fault Zone including Obi there are slices of the ophiolite. Radiometric dates and fossil evidence indicate Mesozoic ages for the ophiolites of the region. Pilot studies using Sm–Nd dating have been carried out on mineral separates from east Halmahera ophiolitic cumulate gabbros and yielded Jurassic ages. Middle-late Jurassic ages of approximately 145 Ma were reported from basic dykes on Gag island by Pieters et al. (1979) and K–Ar analyses of basaltic dykes from Gag carried out during this project gave ages of 166 ± 6 and 142 ± 4 Ma (Baker, 1997). Supriatna and Apandi (1982) reported Upper Jurassic calpionellid-bearing rocks from central Waigeo and during the course of the SE Asia Research Group research in the region we collected lower Cretaceous calpionellid-bearing mudstones associated with the ophiolites from north Waigeo. On Halmahera and Obi the ophiolite is overlain by Upper Cretaceous arc volcanic, volcaniclastic rocks and pelagic sediments (Hall et al., 1988; Ali and Hall, 1995; Agustiyanto, 1996).

5. Ophiolitic rocks sampled for dating and palaeomagnetism

The ophiolite was sampled for palaeomagnetic study on the logging road leading from the village of Ocimaloleo, SW Obi (Fig. 3) and some of the samples were also isotopically dated. Gabbros, dolerites and basalts are well exposed in several areas. A particularly fine exposure of gabbros intruded by pegmatitic gabbros, dolerites and basalts is present along the Air Pati River approximately 8 km north of the Ocimaloleo logging camp. Intrusive relationships are well displayed along a 200 m section. Dolerite and basalt dykes are sub-vertical (dipping at about 70° to the east), laterally continuous and between 0.3 and 0.5 m in width. The dykes show a consistent strike (approximately NE–SW) indicating an extension direction of 140°. On the logging road, about 100 m above the river valley, there are small exposures of dykes with a similar orientation to those intruding the gabbros although the structural continuity between the exposures is uncertain. Good dyke exposures also occur along the Jikodolong–Ricang logging road where they dip steeply to the northwest. In none of these areas was one-way chilling, characteristic of a true sheeted complex, found.

The host rock in Air Pati is a greenish coarse gabbro to dolerite; in most areas it is homogeneous but locally shows faint mineralogical banding particularly around pegmatitic bodies. Grain sizes are 0.5–5 mm; plagioclase (An_{60-85}) is fresh with polysynthetic twinning and makes up 50% of the mode. Sub-ophitic augite originally made up 45% of the rock but is now mainly altered to pale green actinolitic

Table 1
Locations of palaeomagnetic samples and dated samples referred to in the text (Wr: whole rock; Hb: hornblende; Cpx: clinopyroxene)

Sample	Longitude	Latitude	Rock type	Method	Material	Age
OE93	127.6149	−1.6110	Dolerite dyke	Pmag		
OE94	127.6151	−1.6113	Gabbro	Pmag		
OE95	127.6153	−1.6118	Gabbro	Pmag		
OE96	127.6464	−1.6131	Dolerite dyke	Pmag		
OD233	127.5305	−1.5507	Phyric basalt	K–Ar	Wr	103 ± 13
OJ102	127.4713	−1.5181	Hb diorite	K–Ar	Hb	62 ± 2
OJ107	127.4754	−1.4565	Hb diorite	K–Ar	Wr	83 ± 6
OJ108	127.4726	−1.4514	Gabbro	Sm–Nd	Cpx–Wr	207 ± 29
OR191	127.8513	−1.6330	Aphyric basalt	K–Ar	Wr	96 ± 10
OR193	127.8621	−1.6509	Amphibolite	K–Ar	Hb	80 ± 2
OR262	127.3927	−1.1433	Hb cumulate	K–Ar	Hb	100 ± 4
OS6	127.5775	−1.5800	Trondhjemites	K–Ar	Hb	71 ± 2

amphibole. Rounded, dusty relics of pyroxene remain in the centre of grains whose margins are converted to single crystals or aggregates of actinolite. Replacement of pyroxene is complete in the finer grained dolerite. Opaque grains were not observed in the gabbro but trails of very dark green spinels are associated with large patches of amphibole. Light brown sphene (<5%) occurs in the dolerite. Plagioclase compositions and the abundance of actinolite ($Al_2O_3 = 3.2–5$ wt%) suggests lower actinolite greenschist facies (~400°C). Pegmatitic gabbros of similar composition occur as small pods, lenses and discontinuous veins and may represent late-stage, vapour-rich crystallisation products of the host. Dolerites and basalts that intrude the gabbro typically have an equigranular texture although some ophitic patches remain and the basalts are sparsely clinopyroxene-phyric. Basalt dykes show chilled margins at the contact with the microgabbro host indicating a period of cooling between host formation and dyke intrusion. The dolerites and basalts are mineralogically identical. Plagioclase ($An_{50–60}$) makes up 40–60% of the rock. Primary sub-calcic augite (45–55% modal abundance) is mainly replaced by pale green actinolite; the remaining 5% is made up by small opaque cubes. Other secondary minerals are sphene, pumpellyite and epidote, the latter two are found in small veins.

Dykes found on the logging road above the Air Pati River fall into two petrographic groups. The first group is variably altered and identical to those intruding the Air Pati gabbro. Grain sizes vary between basalt and microgabbro, none contain clinopyroxene, and plagioclase varies in composition between An_{65} and albite. A second group of dolerites contains abundant primary clinopyroxene and has intersertal, quenched or supercooled textures. Clinopyroxenes have elongate and skeletal morphologies; some sub-ophitically enclose plagioclase and infill interstices between plagioclase laths. Microprobe determinations (Agustiyanto 1996) indicate ferroan diopside compositions containing up to 1.34 wt% TiO_2 (typically 0.5–0.7 wt%). Plagioclases are dusty brown, elongate, and vary from calcic (An_{80}) to sodic reflecting variable alteration. Rare orthoclase was identified in some samples by microprobe. Groundmass glass is now altered to very pale green chlorite; Fe–Ti oxides are deep red spinels which have euhedral to subhedral morphology and relatively large grain sizes suggestive of early crystallisation. The occurrence of rare chlorite and ?smectite aggregates suggest alteration of olivine phenocrysts, a mineral not found in other high level crustal rocks from the region.

In general, metamorphic assemblages are characteristic of sea floor metamorphism; local temperatures up to 500°C are indicated but pressures are typically low. Mineralogical changes in the dolerites indicate metamorphism at temperatures >250°C at low pressures (Baker, 1997), mainly under conditions corresponding to the prehnite–pumpellyite to prehnite–actinolite facies of Liou et al. (1987). There are several varieties of gabbros found in Obi, including olivine gabbros, gabbro–norites, and hornblende gabbros and many have cumulate textures. They are generally very fresh and contain cumulus pyroxenes and plagioclase, with intercumulus amphibole. Like the dolerites, some gabbros shown signs of metamorphism under low-grade metamorphic conditions, but between the pumpellyite–actinolite and epidote–actinolite facies. All this is typical of submarine hydrothermal metamorphism at mid-oceanic ridges (Yardley, 1989). Thus it was hoped that, if these rocks could be dated isotopically, the ages obtained would indicate either the date of primary igneous crystallisation from least altered samples, or the age of subsea floor metamorphism which is likely to have occurred soon after magmatism.

6. Age constraints on the ophiolitic rocks from Obi

Pillow lavas forming part of the ophiolite are well exposed in several localities on the islands of the Obi group but nowhere have we found the closely associated sedimentary rocks, such as cherts and pelagic limestones, to contain fossils that could be dated.

Isotopic dating by different methods was carried out using

a variety of ophiolitic samples selected on petrographic criteria and judged to be as fresh as possible. K–Ar dating was performed on ophiolitic dolerites and gabbros from the upper crustal sections of the Obi Ophiolite Complex (Table 1). The K–Ar ages obtained range between 103 and 27 Ma. Dolerite dyke samples OR191 and OD233 yielded the oldest ages. Sample OR191 was collected on the Bobo logging road in SE Obi. OD233 was collected on the logging road from Ricang to Jikodolong in central Obi. The ages are 96 ± 10 Ma (OR191) and 103 ± 13 Ma (OD233) which are within error of one another and are interpreted as minimum ages. This is based on the premise that the low-grade metamorphism suffered by these rocks is likely to have led to radiogenic argon loss, not argon gain. Therefore although the apparent ages may not accurately date a geological event (such as cessation of sub-sea floor metamorphism) we suggest they can be used as indicators of a minimum age for the ophiolitic rocks.

Plutonic rocks intrude the ophiolite and fresh hornblende separates from two hornblende diorites (OJ107, OJ102) and one trondhjemite (OS6) yield Cretaceous ages of 83 ± 6, 62 ± 2 and 71 ± 2 Ma (Baker, 1997). $^{40}Ar/^{39}Ar$ dating of hornblendes from similar diorites on Halmahera (Ballantyne, 1990) indicates two phases of late Cretaceous arc-related igneous activity (94–80 Ma: Cenomanian–Campanian and 75–72 Ma: Campanian–Maastrichtian).

In parts of the island the ophiolite includes cumulate gabbros and norites, and extensive areas of serpentinised peridotite, locally with a thick laterite cover, representing its deeper parts. Clinopyroxene and plagioclase from several cumulate gabbros were separated in an attempt to determine the age of ophiolite formation using the Sm–Nd technique. Sm–Nd ages from fresh ophiolitic gabbros underlying the dolerites are variable due to analytical difficulties, specifically the accurate determination of the low radiogenic Nd contents. Only one (OJ108), an olivine gabbro collected on the logging road from Ricang to Jikodolong, yielded an age from a two point isochron using clinopyroxene and whole rock analyses which is 207 ± 29 Ma. This is interpreted as indicating that the ophiolitic rocks on Obi may be as old as early Jurassic. This age is consistent with early–Middle Jurassic Sm–Nd (197 ± 6 and 152 ± 20 Ma) and the K–Ar (166 ± 6 Ma) isotopic ages obtained from ophiolitic rocks from nearby Halmahera and Gag, respectively (Baker, 1997). It is also consistent with the presence of calpionellids in sedimentary rocks on Waigeo associated with ophiolites that are part of the same ophiolitic province.

Upper Cretaceous volcaniclastic rocks and pelagic limestones of the Leleobasso Formation rest unconformably on the ophiolite (Ali and Hall, 1995; Agustiyanto, 1996) and are Campanian to Maastrichtian based on foraminifera. Stratigraphic (Agustiyanto, 1996) and isotopic data (Forde, 1997) from rocks post-dating the ophiolite indicate that it is older than late Cretaceous. From stratigraphic arguments summarised above it is clear that the ages younger than late Cretaceous must be partly or completely reset but it was hoped that the older ages might indicate the age of ophiolite formation. The exact age of formation of the ophiolite remains uncertain due to factors such as lack of suitable mineral phases for dating, low K contents and metamorphism. The possibility that these rocks contain excess ^{40}Ar (leading to older ages) cannot be ruled out although sub-sea floor metamorphism would be expected to release all previously acquired argon from a rock with a dolerite mineralogy suggesting that the oldest ages represent reliable minima. Younger K–Ar ages from ophiolitic rocks are interpreted to be the result of local resetting due to late Cretaceous and Tertiary arc magmatism and related thermal events. We recognise that the K–Ar dates are inadequate to reliably date the ophiolite but at present we have no better data to reliably indicate its true age. Based on the isotopic and stratigraphic data the ophiolitic dolerites and gabbros are early Cretaceous or older.

7. Palaeomagnetism

Sites were located to ±30 m using a Magellan Navpro 1000 GPS receiver. Specimens were obtained using a gasoline powered rock-drill which was used to cut 25 mm diameter mini-cores. The cores were oriented to ±2° using a magnetic compass inclinometer. Six to eight oriented mini-cores were collected from each site. The structural attitude was measured at each site to provide a tilt-correction; the orientation of the inclined dykes was measured and later used to correct the magnetic vectors to their original, presumed, vertical orientation. All samples were taken from dykes, and where they intruded layered microgabbros they cut the layering at a high angle. Stability of the natural remanent magnetisation (NRM) of each specimen was assessed after stepwise alternating field demagnetisation (AF) was used to isolate the various magnetisation components held within the rock. The specimens were analysed using a 'Molspin' spinner magnetometer in tandem with a 'Molspin' demagnetiser. Examples of demagnetisation vector end point plots (Zijderveld, 1967) are shown in Fig. 4.

7.1. NRM characteristics

Sites OE93-95 (Table 1) were sampled from a small area of continuous excellent exposure in the Air Pati river approximately 13 km from Ocimaloleo. Site OE93 was sampled from an approximately 0.5 m wide fine grained dyke intruding a microgabbro. Initial NRM intensities for specimens from this site show wide range of values (20–140 mA/m). The majority of specimens from this site (e.g. Fig. 4a) carry a low coercivity magnetisation (removed at 5–10 mT) which, prior to restoring the dykes to the palaeovertical, is parallel to the present geomagnetic field direction (i.e. it is a viscous remanence).

Sites OE94 and OE95 are from the microgabbro and were sampled 1.0 m east and 1.5 m west of the OE93 dyke,

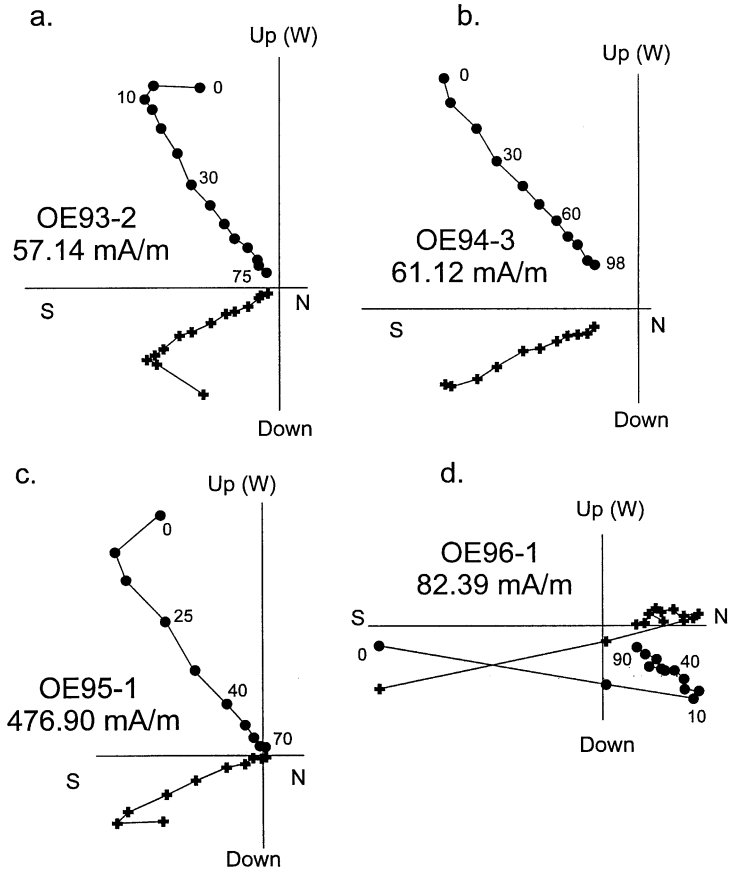


Fig. 4. Examples of AF demagnetisation vector end point (Zijderveld, 1967) plots for the Obi ophiolite sites. Filled circles/crosses represent the remanence vector on the horizontal/vertical (N–S oriented) plane. Numbers indicate the applied demagnetisation field (mT). The initial NRM intensity is given in milliamperes per metre (mA/m).

respectively. Although the two sites are from the same unit, they exhibit notably different demagnetisation behaviour (Figs. 4b and c) and have different palaeomagnetic characteristics, which suggests a slightly different crystallisation-cooling history for the magnetic grains in the two sites. All of the specimens from site OE94 carry an essentially single component remanence (Fig. 4b). NRM intensities vary between 15 and 65 mA/m, and median destructive fields are 40–60 mT. Many of the specimens from site OE95 carry, in addition to a high stability component, a viscous remanence that is removed at 5–10 mT (Fig. 4c). NRM intensities for this site vary between 150 and 740 mA/m, an order of magnitude increase on site OE94. Also, median destructive fields are less than in the site OE94 samples, with values of 25–30 mT.

Site OE96 was sampled from a sheeted dyke sequence

Table 2
Summary of palaeomagnetic data. N = Number of specimens. NRM = initial intensity in mA/m. IRM ratio = IRM at 0.3 T/IRM at 0.86 T. Peak IRM expressed in mAm^2. F = Fisher (1953) statistics used to calculate mean direction at site level

Site	Unit	N	In situ		Dyke correction	Tilt corrected		α_{95}	k	NRM range	IRM ratio	Peak IRM
			Dec	Inc		Dec	Inc					
OE93	dyke	6	233.1	39.4	144/23	233.3	16.4	5.3	162.1	15–70	1.00	18,318
OE94	gabbro	6	227.0	37.8	144/23	228.3	14.9	4.2	261.6	20–70	0.99	850
OE95	gabbro	6	225.9	33.8	144/23	227.1	11.0	3.3	396.2	150–740	0.99	154,357
Mean		3	228.6	37.0	(144/23)	229.5	14.1	6.4	366.9			
OE96	dyke	6	206.1	8.9	210/20	209.4	9.7	7.7	76.7	70–205	0.99	107,261
								Angular separation				
OE93–96	2	216.1	23.3					34.7				
						219.4	12.1	20.1				

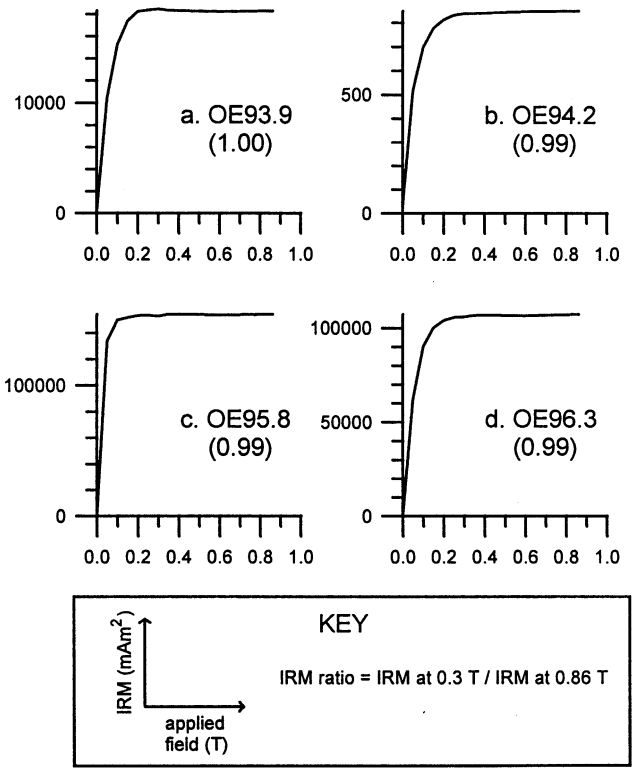


Fig. 5. IRM acquisition curves for representative samples from the west Obi ophiolite sites. In all cases, the IRM saturates in fields between 0.2 and 0.3 T. This behaviour suggest that for many samples the remanence is carried by magnetite. IRM ratio and peak IRM values are listed in Table 1.

next to a disused logging road about 12 km from Ocimaloleo (Table 1). NRM intensities vary between 80 and 200 mA/m. Demagnetisation indicates that a large component (>50%) of this is due to a viscous remanence. Beyond 10 mT, directions are stable. It is worth noting that a single specimen from this site (OE96.1, Fig. 4d) carries a normal polarity remanence with a direction antipodal to the reverse polarity high-stability component identified in the other specimens from this site.

7.2. Isothermal remanent magnetisation experiments

Isothermal remanent magnetisation (IRM) analysis was carried out on one specimen from each of the four sites to provide basic information on the magnetic carriers. The IRM was generated using a 'Molspin' pulse magnetiser with a peak direct field of 0.86 T. The IRM was measured between steps using a 'Molspin' spinner magnetometer. The shape of the IRM curve (as well as the peak IRM value) was used to evaluate the characteristic remanence carrier(s). The IRM ratio (Ali, 1989: the ratio of the IRM at 0.3 T/IRM at 0.86 T), provides a simple numerical method of describing the IRM curve. Specimens with IRM ratios approaching 1.0 effectively saturate in low direct fields, suggesting that the remanence is low coercivity carrier such as magnetite. In cases where specimens do not saturate at low fields (say when the IRM ratio is less than 0.9), then it is likely that the remanence is carried by a higher coercivity mineral. Data from the analysed specimens are presented in Table 2, and are summarised in Fig. 5. All of the specimens have IRM ratios of greater than 0.98 suggesting that the remanence of the Obi ophiolite sites is carried by magnetite.

7.3. NRM/IRM demagnetisation

As well as the standard directional and magneto-mineralogical studies, the NRM/IRM demagnetisation technique (Fuller et al., 1988; Cisowski et al., 1990) was applied to a representative specimen from each site. The method, based

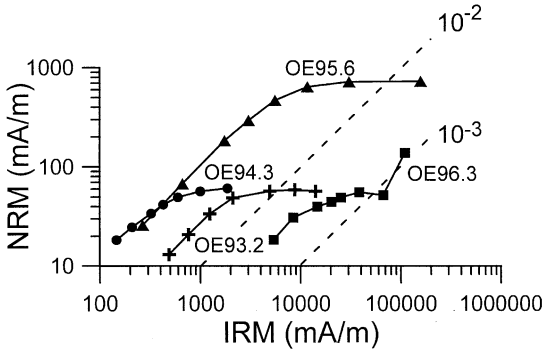


Fig. 6. NRM/IRM demagnetisation curves for representative specimens from each site.

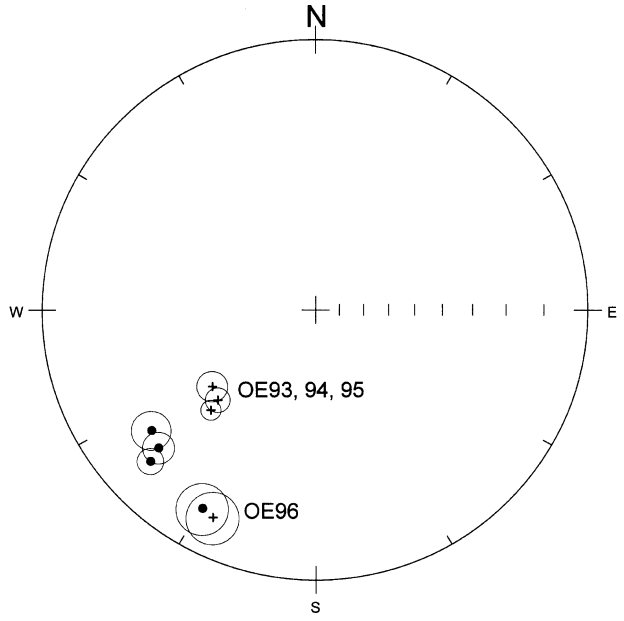


Fig. 7. Stereographic plot showing the in situ (crosses) and vertically restored (filled circles) Obi ophiolite site mean data. The vectors are all downward dipping and shown with their 95% confidence circle.

on empirical observations, is used to determine whether an igneous body has a primary thermoremanent magnetisation (TRM), or a secondary chemical remanent magnetisation (CRM). In this test, the decay of a specimen's NRM during AF demagnetisation is compared with the decay of its IRM at equivalent fields. According to Fuller et al. (1988); Cisowski et al. (1990), if the NRM/IRM ratio for most of the demagnetisation steps is greater than 10^{-2} then the NRM is likely to be a TRM. However, when the ratio is less than 10^{-3}, the remanence is probably the result of a secondary CRM.

Specimens from sites OE93-95 (Fig. 6) have NRM/IRM ratios in excess of 10^{-2} which suggests that the remanence is primary. Specimen OE96.3 has a more 'jumpy' curve for the first three demagnetisation steps (due to the large VRM it carries). For demagnetisation steps above 22.5 mT, the NRM/IRM ratios are greater than 2×10^{-3} with the latter steps typically $2.5^{-3} \times 10^{-3}$. Thus, OE96.3 falls somewhere between a clear primary TRM and a clear secondary CRM. We assume that the remanence of site OE96 is primary; the large southerly declination deflection and relatively simple demagnetisation behaviour following removal of the VRM, suggests that the remanence cannot be a recent CRM.

7.4. Site mean directions

Characteristic components of magnetisation for each specimen were identified from Zijderveld (1967) plots, and calculated using a Core Magnetics software package that uses Kirschvink's (1980) principal component analysis. Site mean directions (Fig. 7 and Table 2) were calculated using the statistics of Fisher (1953). In calculating the mean in situ and tilt corrected directions for the Obi ophiolite, sites OE93-95 have been grouped to generate an outcrop-mean. Calculating the mean directions for this group and site OE96 yields an in situ value of $D = 216.1°$, $I = 23.3°$, where the angular separation is 34.7°. Restoring the two dyke outcrops to the palaeo-vertical results in a mean direction $D = 219.4°$, $I = 12.1°$, and brings the vectors closer together such that the angular separation is 20.1°.

7.5. Latitude of formation and regional implications

If the magnetisation of the Obi ophiolite is primary ($D = 219.4°$, $I = 12.1°$) then it must have been acquired at a subequatorial latitude. It is not possible to discriminate between the northern or southern hemisphere because of analytical precision and the complex Cenozoic rotation history of the Obi region. Tectonically, the Obi ophiolite is now within the Sorong Fault Zone and may therefore have undergone relatively recent CW and/or CCW rotation (Ali and Hall, 1995) since it was separated from the main plate at some time in the late Neogene by a splay of the Sorong Fault. Assuming it formed part of the Philippine Sea Plate, prior to its separation it must also have experienced up to 40° clockwise rotation in the Neogene and about 50° clockwise rotation between the middle Eocene and Oligocene.

8. Conclusions

In recent years palaeomagnetic data have been obtained from many formations which formed on crust within the Philippine Sea Plate (Haston and Fuller, 1991; Ali and Hall, 1995; Hall et al., 1995a). The largest subset of data is from the late Paleogene arc rocks that formed along the southern boundary of the Philippine Sea Plate. This arc was generated in response to subduction of the oceanic crust north of the Australian continent as the Indo-Australia plate moved towards the equator. Inclination data from the arc sequence suggest that the southern edge of the Philippine Sea Plate was at 12–15°S at the time of arc-continent collision at about 25 Ma (Hall et al., 1995a). Data from older rocks indicate that this part of the plate had been closer to the equator during the early Cenozoic and the Late Cretaceous. The new data from the Obi ophiolite presented here indicate that possibly as long ago as the early–Middle Jurassic this part of the plate also occupied a sub-equatorial latitude. Whether it underwent appreciable latitudinal motion between that time and the late Cretaceous is uncertain. The ophiolite was close to the equator at all times for which we have data: this includes all the Cenozoic and Cretaceous Philippine Sea Plate sites from the region. The plate may also have undergone significant longitudinal motion since the ophiolite formed.

The long history of the Philippine Sea Plate, and its central location within the western Pacific convergence zone, suggest that it has played an important role in the

tectonic development of the region. Unfortunately, the palaeomagnetic database for rocks formed prior to the middle Eocene is very small and modelling the basic plate framework for this period is difficult. However this situation could be redressed through the study of the older Philippine Sea Plate rocks that may have been detached from the plate in northern New Guinea and the Philippines, as well as other localities within the Sorong Fault Zone. Magnetic inclinations would provide much valuable information and it might be possible to unravel potentially complex declination histories based on our knowledge of the Cenozoic rotation of the main plate, particularly its older history.

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